AQI Prediction System for Karachi, Islamabad, and Lahore

Executive Summary

This project develops an Air Quality Index (AQI) prediction system for three major Pakistani cities: Karachi, Islamabad, and Lahore. The system fetches historical and forecast data from Open-Meteo, applies feature engineering, trains machine learning models (Ridge, RandomForest, GradientBoosting, XGBoost), and generates 72-hour AQI forecasts with SHAP analysis for interpretability. Daily backfilling updates the historical dataset, and a Streamlit UI visualizes forecasts, SHAP plots, EDA, and health recommendations. The approach focuses on time-series data handling, city-specific modeling, and robustness against Open-Meteo's unreliable AQ forecasts. Key challenges include data misalignment (negative R^2 in tests) and feature consistency, addressed through imputation and retraining.

Introduction

Project Background

Air quality monitoring is critical for urban health in Pakistan, where cities like Lahore face severe smog and Karachi experiences industrial pollution. This project builds an automated AQI prediction system to forecast 72-hour AQI using Open-Meteo API data. The system supports daily data updates, model retraining, and an interactive UI for visualization.

Objectives

- Fetch and process historical and forecast data from Open-Meteo.
- Engineer features for time-series modeling (lagged, rolling, cyclic, interaction).
- Train city-specific models and evaluate performance.
- Generate forecasts with SHAP for explainability.
- Backfill daily data and automate workflows via GitHub Actions.
- Provide a Streamlit UI for forecasts, SHAP, EDA, and health recommendations.

Scope

- Cities: Karachi, Islamabad, Lahore.
- Data: Hourly AQ (US AQI, PM2.5, PM10, O3, CO, NO2, SO2) and weather (temperature, humidity, wind, cloud cover, precipitation).
- Models: Ridge (for Karachi), XGBoost (for Islamabad/Lahore), RandomForest, GradientBoosting.
- Automation: Daily backfill/training, 3-hourly forecasting.

Approach

Data Acquisition

- **Historical Data**: Fetched 82 days of AQ and weather data using `fetch_historical_data.py`. Merged hourly data, validated (e.g., temperature 0–50°C), and saved to `historical_combined_cities_new.csv`.
- **Forecast Data**: Fetched 72-hour weather and AQ forecasts using `forecast_aqi.py`. Used AQ forecasts for lagged/rolling features, historical values for direct AQ features.
- **Backfilling**: `backfill_data.py` fetches 24-hour data daily, appends to historical CSV.

Feature Engineering

- **Temporal**: `dayofweek`, `hour`, `month`, `hour_sin`, `hour_cos`, `month_sin`, `month_cos`.
- **Lagged**: `pm2_5_lag_1`, `pm2_5_lag_6`, `pm2_5_lag_24`, `pm10_lag_1`, `pm10_lag_6`, `pm10_lag_24`.
- **Rolling**: `pm2_5_rolling_mean_24h`, `pm2_5_rolling_std_24h`, `pm10_rolling_mean_24h`, `pm10_rolling_std_24h`.
- **Interaction**: `temp_humidity_interaction` (temperature × humidity), `wind_pm25_interaction` (wind speed × PM2.5).
- **Imputation**: Forward-fill and zero-fill for NaNs, historical values for AQ features in forecasts.

Model Selection and Training

- **Models**: Ridge (linear with regularization), RandomForest (ensemble trees), GradientBoosting (sequential boosting), XGBoost (optimized boosting).
- **Training Script**: `model_testing_new.py` performs city-specific training on 80% historical data, with time-series cross-validation (5 splits). Loads saved models/scalers if available for retraining on full data.
- **Retraining**: Fine-tunes saved models on updated historical data (including backfilled 24-hour data) to incorporate new patterns.

Forecasting

- **Script**: `forecast_aqi.py` fetches weather and AQ forecasts, adds features, predicts AQI using XGBoost (or Ridge for Karachi), and performs SHAP analysis.
- **Hybrid Approach**: Uses Open-Meteo AQ forecasts for lagged/rolling features, historical values for direct AQ features to mitigate unreliability.
- **SHAP**: Generates summary plots for feature importance (e.g., `pm2_5`, `temp_humidity_interaction`), saved to `data/shap_<city>.png`.

Automation

- **GitHub Actions**: `workflow.yml` runs:
  - Daily (00:00 UTC): Backfill (backfill_data_new.py) and retraining (`model_testing_new.py`).
  - Every 3 hours: Forecasting (`forecast_aqi.py`).

Results

Training Performance

City	Model	MAE	RMSE	R^2	CV_RMSE
Islamabad	XGBoost	6.52	12.13	0.82	16.06
Islamabad	GradientBoosting	6.88	12.32	0.81	15.95
Islamabad	Ridge	10.16	13.43	0.77	25.84
Islamabad	RandomForest	7.50	13.55	0.77	16.96
Karachi	Ridge	1.27	1.89	0.87	4.98

Karachi	XGBoost	1.43	5.06	0.04	3.80
Karachi	RandomForest	4.59	11.51	-3.96	2.83
Karachi	GradientBoosting	4.57	11.59	-4.03	2.57
Lahore	XGBoost	3.71	6.16	0.88	21.30
Lahore	GradientBoosting	4.10	6.96	0.85	21.46
Lahore	RandomForest	4.37	7.69	0.81	20.81
Lahore	Ridge	9.28	11.38	0.59	40.55

SHAP Insights

- Karachi (Ridge): Top features include `pm2_5`, `pm10`, `temp_humidity_interaction`.
- Islamabad/Lahore (XGBoost): Top features include `pm2_5_rolling_mean_24h`, `pm10_lag_24`, `wind_pm25_interaction`.

Conclusion

This project successfully implements an AQI prediction system with automated data fetching, training, forecasting, and visualization. The hybrid approach of using historical AQ for direct features and Open-Meteo forecasts for temporal features mitigates data unreliability, achieving low training errors. Future improvements include LSTM integration for temporal modeling and incorporating local AQ sources (e.g., Pakistan EPA). The system is deployable via GitHub Actions for production use.